

Limitations of Vamsi et al. (2022)

Reference

Vamsi, K. R. et al. *Prediction of Micro Vascular and Macro Vascular Complications in Type-2 Diabetic Patients using Machine Learning Techniques*. International Journal of Advanced Computer Science and Applications (IJACSA), Vol. 13, No. 11, 2022.

Identified Limitations

1. Modular and Independent Prediction Architecture

2.

The proposed system models each diabetes-related complication. nephropathy, neuropathy, retinopathy, cardiovascular disease, and peripheral vascular disease. as a separate classification module. This modular design implicitly assumes independence between complications and does not incorporate shared biological mechanisms that drive comorbidity. As a result, the progression or predicted risk of one complication does not inform the prediction of others, despite well-established clinical interdependence.

2. Absence of Shared Patient Representation Learning

The framework relies on classical machine learning algorithms, including Random Forests, Decision Trees, Logistic Regression, and AdaBoost. Each model independently learns feature relevance from the same clinical variables, such as HbA1c, BMI, blood pressure, and duration of diabetes. This prevents the learning of a unified latent representation capturing systemic metabolic dysfunction across organs and limits the model's ability to generalize higher-order physiological patterns.

3. Visualization-Only Use of Comorbidity Relationships

Although graph-theoretic methods are employed to visualize relationships between complications, these relationships are not incorporated into the predictive pipeline. The comorbidity network serves an exploratory role only and does not mathematically influence risk estimation. Consequently, identified inter-complication relationships remain disconnected from the model's decision logic.

4. Static Risk Categorization Without Decision Support

Model outputs are restricted to static categorical risk levels. Low, Medium, and High. intended for clinician interpretation. The system does not provide guidance on clinical actions, treatment escalation, or guideline-based interventions. This creates a decision gap between risk estimation and evidence-based care, limiting the framework's utility as a clinical decision-support system.

5. Limited Explainability

Explainability is provided through global feature importance scores aggregated at the population level. The framework lacks instance-level explanations and does not analyze feature interactions that jointly influence risk across multiple complications. As a result, clinicians are unable to understand patient-specific risk drivers or synergistic effects between clinical variables.

6. Evaluation Restricted to Independent Binary Metrics

Each complication is evaluated using independent binary classification metrics, such as accuracy and AUROC. The study does not assess joint prediction behavior across complications using multi-label evaluation measures. This limits insight into how well the system captures co-occurring complication risk at the patient level.

Role of Vamsi et al. (2022) in the Present Study

This work serves as the **primary dataset source and baseline modeling reference** for subsequent research. While it establishes the feasibility of machine learning-based risk prediction for diabetes complications, its architectural and methodological limitations motivate the development of relationship-aware, multi-label frameworks capable of unified representation learning, explicit comorbidity modeling, and clinically actionable decision support.

Limitations of Nicolucci et al. (2022)

Reference

Nicolucci, A. et al. *Prediction of complications of type 2 Diabetes: A Machine Learning approach. Diabetes Research and Clinical Practice*, 190 (2022) 110013.

Identified Limitations

1. Independent Modeling of Each Complication

The study employs six separate XGBoost models to predict individual diabetes complications. nephropathy, neuropathy, retinopathy, cardiovascular disease, cerebrovascular disease, and peripheral vascular disease. While this approach allows optimization for each outcome independently, it does not model comorbidity structure or shared disease pathways. The presence or predicted risk of one complication does not influence others, despite strong clinical evidence that these complications frequently co-occur and progress interdependently.

2. Absence of Explicit Feature Interaction Analysis

Although the authors acknowledge that diabetes complications arise from the interaction of multiple clinical factors, the modeling framework does not provide feature interaction explanations. Explainability is limited to feature importance rankings, without SHAP interaction values or joint effect analysis. Consequently, the model cannot reveal how combinations of variables. such as duration of diabetes and blood pressure. synergistically elevate risk across organs.

3. Lack of Integrated Clinical Decision Support

The system focuses exclusively on risk prediction and does not extend to treatment guidance or clinical recommendations. Despite discussing the clinical benefits of interventions such as SGLT2 inhibitors, these insights are not operationalized within the model. This limits the framework's ability to support real-world clinical decision-making beyond probabilistic risk estimation.

4. Sensitivity–Specificity Trade-off with Clinical Implications

Several models prioritize sensitivity at the expense of specificity. For example, cardiovascular disease prediction achieved 70.5% sensitivity but only 59.2% specificity, while cerebrovascular disease achieved 89.1% sensitivity with 52.6% specificity. Such imbalances increase false-positive rates, potentially leading to unnecessary clinical investigations and reduced trust in automated risk stratification tools.

5. Variable External Validation Performance

External validation across multiple centers revealed substantial variability in model performance. For instance, cerebrovascular disease prediction AUROC ranged from 0.59 to 0.86 depending on the center. The study does not provide a systematic analysis of this performance degradation, such as population heterogeneity, data distribution shifts, or feature availability differences.

6. Evaluation Limited to Binary Classification Metrics

Model evaluation is restricted to binary performance metrics for each complication independently. The framework does not assess joint prediction performance using multi-label evaluation measures, despite predicting multiple complications per patient. This limits insight into how effectively the system captures concurrent complication risk in real-world clinical populations.

Role of Nicolucci et al. (2022) in the Present Study

This work represents an important step toward clinically validated machine learning models for diabetes complication prediction. However, its reliance on independent models, absence of interaction-level explainability, and lack of decision-support integration highlight the need for unified, relationship-aware architectures that jointly model comorbidity, enhance interpretability, and bridge the gap between prediction and clinical action.